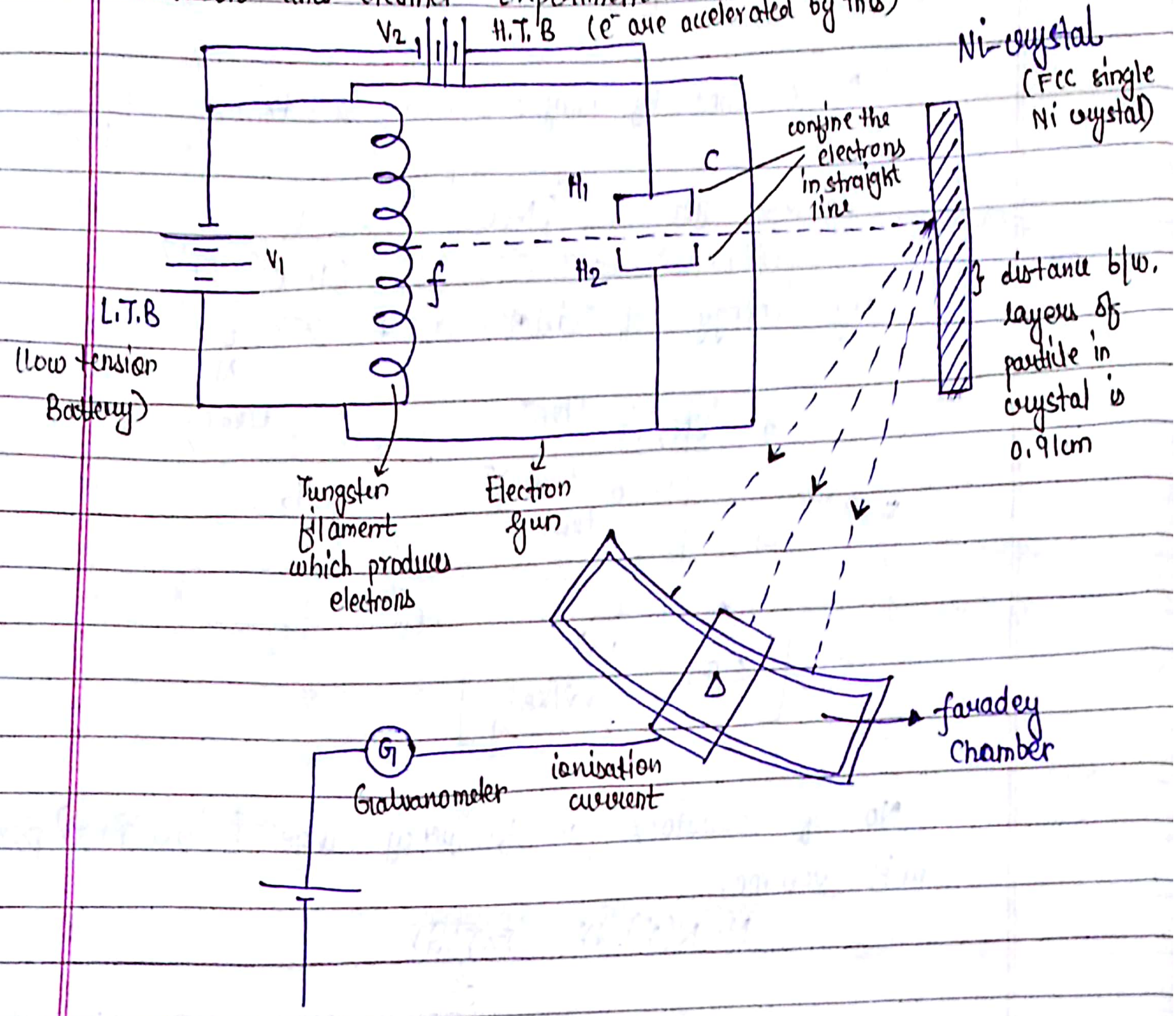
Page No.
Date : 5/03/18

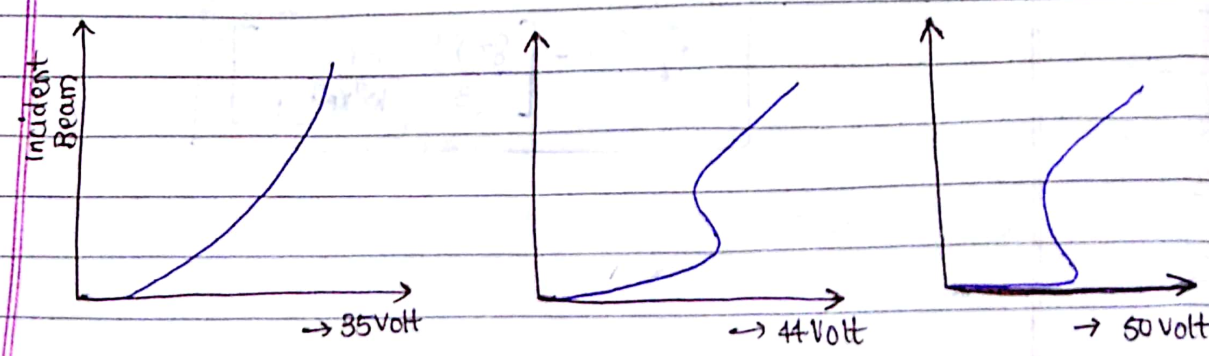
5/03/18

Wave - Particle Duality

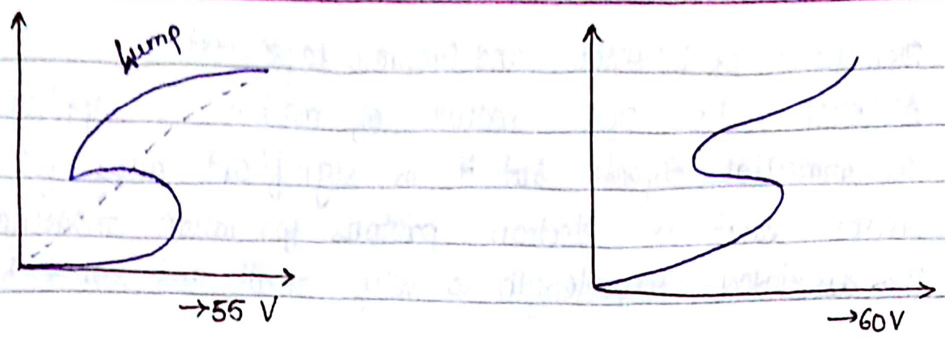
Davisson and Germer Experiment :



35V — 60volt : H.T.B

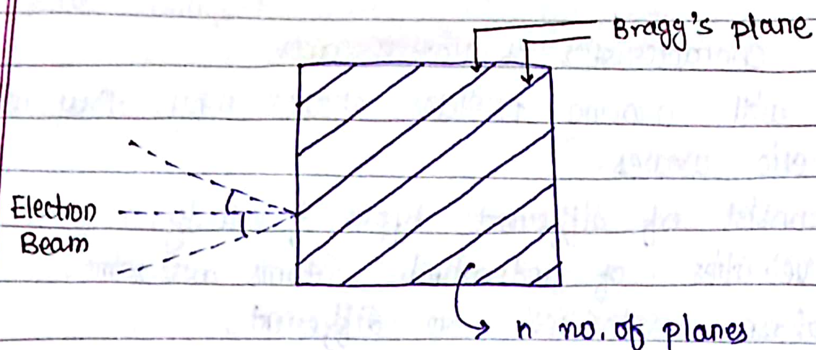


varies linearly



wave - shape
(electrons behave like a wave)

$\lambda = \frac{12.28}{\sqrt{V}} Å = \frac{12.28}{\sqrt{55}} = 1.66 Å$



Bragg's Law :

When light is incident on a crystal (single or polycrystalline) then it gives diffraction inside it.

$$\boxed{2d\sin\theta = n\lambda}$$

d → distance b/w. 2 consecutive planes
n → order of deflection

For first order deflection : n=1

$$2(0.91)\sin 65° = \lambda$$

$$\text{Wavelength}, \lambda = 1.65 Å$$

- Davisson and Germer Experiment proves the existence of moving particle i.e. electron.

$$\boxed{\underbrace{\frac{1}{\lambda}}_{\text{wavelength}} = \underbrace{\frac{n}{2d\sin\theta}}_{\text{Bragg's Law}} = \underbrace{\frac{p}{h}}_{\text{Debroglie Relation}} = \underbrace{\frac{\sqrt{2mE}}{h}}_{\text{Acceleration through Voltage}}}$$

E → electric field

Page No.
Date : / /

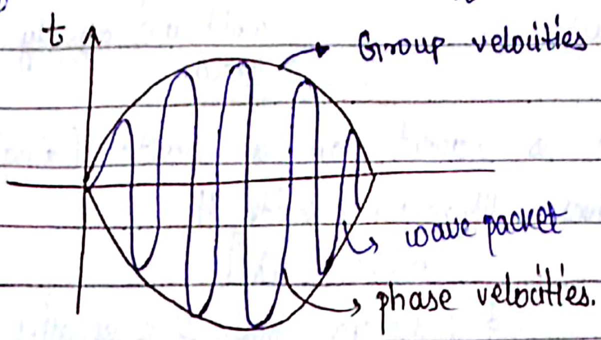
Significance of Davisson and Germer Experiment :-

- Although the Dual nature of matter is applicable to all material objects but it is significant only for microscopic bodies such as electron protons for large material object.
- The associated wavelength is very small and can't be measured.

Wavelength is very high ⟶ microscopic
Wavelength is very small ⟶ macroscopic

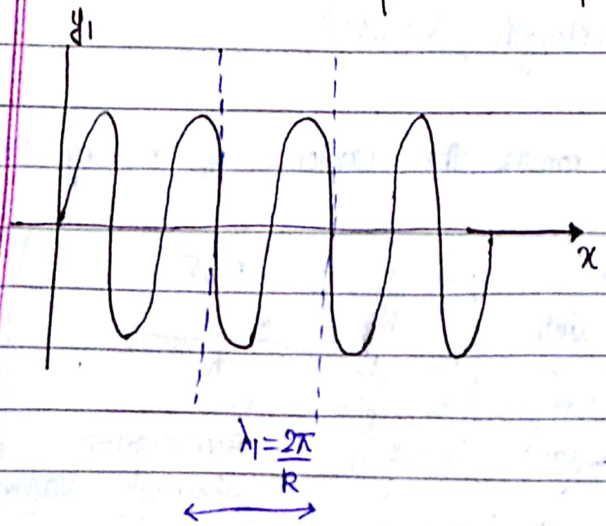
Characteristics of Matter Waves.

1) Associated with moving particles, that's why they are not electromagnetic waves.
2) Atoms consist of different types of velocity.
Group velocities of individual atoms are same. while phase velocities are different.



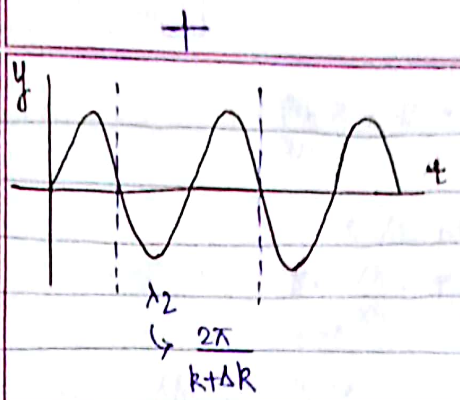
3) It is based on the uncertainity principle and De-Broglie hypothesis.

03/18 Wave packets Group Velocity and Phase Velocity

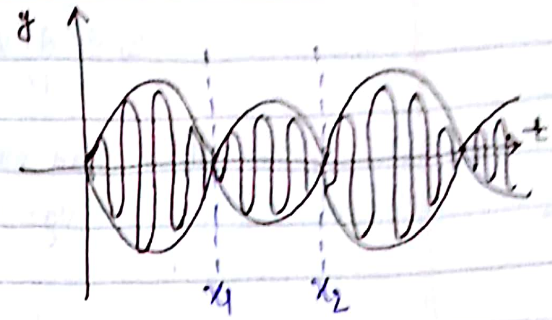


When 2 or more waves have slightly diff λ, ν, amplitude & Intensity superimpose on each other and the resultant produced is the combination of phase and group velocity.

Page No.
Date : / /



$\Rightarrow$



Group velocity → combination of 'n' no. of wave's velocity
phase velocity → velocity of individual particle in monochromatic light.

$y_1 = A\sin(\omega t - kx)$ —①

$y_2 = A\sin[(\omega+\Delta\omega)t - (k+\Delta k)x]$ —②

$y = y_1 + y_2$

$y = A\sin[(\omega t - kx)] + A\sin[(\omega+\Delta\omega)t - (k+\Delta k)x]$ —③

$y = 2A.\cos\left[\left(\frac{\Delta\omega}{2}\right)t - \left(\frac{\Delta k}{2}\right)x\right] . \sin\left[\left(\frac{2\omega+\Delta\omega}{2}\right)t - \left(\frac{2k+\Delta k}{2}\right)x\right]$

Since $\Delta\omega$ and Δk $\Rightarrow$ $2\omega + \Delta\omega \approx 2\omega$; $2k + \Delta k \approx 2k$

$y = 2A \cos\left[\left(\frac{\Delta\omega}{2}\right)t - \left(\frac{\Delta k}{2}\right)x\right] . \sin(\omega t - kx)$

Resultant $R = 2A\cos\left(\frac{\Delta\omega}{2}\right)t - \left(\frac{\Delta k}{2}\right)x$

$x \to 0$ and $t \to 0$;

$R = 2A$

$R_{max} = 2A$

Group Velocity, $v_g = \lim_{\Delta k \to 0} \frac{\left(\frac{\Delta\omega}{2}\right)}{\left(\frac{\Delta k}{2}\right)} = \lim_{\Delta k \to 0}\left(\frac{\Delta\omega}{\Delta k}\right)$

$$\boxed{v_g = \frac{d\omega}{dk}}$$

Phase velocity, $v_p = \frac{\omega}{k}$

Relation b/w. v_g and v_p $\Rightarrow$ $v_g = \frac{d\omega}{dk}$; $v_p = \frac{\omega}{k}$

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Page No.
Date : / /

$$V_g = \frac{d(V_p . K)}{dK} = V_p + K\frac{dV_p}{dK}$$

∴ Divide and multiply by $d\lambda$ ⇒

$$V_g = V_p + \left(\frac{dV_p}{d\lambda} \cdot \frac{d\lambda}{dK}\right) \cdot K$$

$$\left(\therefore K = \frac{2\pi}{\lambda} \; ; \; \lambda = \frac{2\pi}{K} \; ; \; \frac{d\lambda}{dK} = \frac{-2\pi}{K^2} \right)$$

$$V_g = V_p + \frac{dV_p}{d\lambda} \cdot \left(\frac{-2\pi}{K}\right)$$

$$V_g = V_p - \frac{2\pi}{K} \cdot \frac{dV_p}{d\lambda}$$

$$\boxed{V_g = V_p - \lambda \cdot \frac{dV_p}{d\lambda}}$$ → Relation b/w. group velocity and phase velocity.

For a dispersive Medium :
Variation of wave vector K wrt to wavelength and ν is called dispersive medium.
When the dispersion (the scattering) ⇒ $V_g = V_p$

A medium in which wave velocity is frequency or wavelength independent then medium is called Dispersion Medium.

Relation b/w. Group Velocity and Particle velocity :

$V_g = V_{particle}$

$\omega = 2\pi\nu$

$\omega = \frac{2\pi E}{h}$

$K = \frac{2\pi}{\lambda} \Rightarrow K = \frac{2\pi}{h} \cdot p$

$V_g = \frac{d\omega}{dK}$ — ①

wave vector K.

$d\omega = \frac{2\pi}{h} dE$

$dK = \frac{2\pi}{h} \cdot dp$

$$V_g = \frac{\frac{2\pi}{h} \cdot dE}{\frac{2\pi}{h} \cdot dp} = \frac{dE}{dp} \quad \text{— ②}$$

Page No.
Date: / /

$$E = \frac{1}{2}mv^2 = \frac{1}{2}\frac{m^2v^2}{m} = \frac{p^2}{2m}.$$

$dE = \frac{p.dp}{m}$ → momentum of a moving e^- or a particle.

putting in eq. ②:

$$\boxed{v_g = \frac{p}{m} = v_{particle}}$$

Relation b/w. v_g, v_p and c.

Group velocity, phase velocity → can't be greater than velocity of light.

c ↳ velocity of light

$$v_g = \frac{d\omega}{dk} \quad \text{and} \quad v_p = \omega/k$$

$$v_{particle} = v_g = \frac{d\omega}{dk} \Rightarrow v_p = \frac{2\pi E/h}{2\pi p/h} = \frac{E}{p} = \frac{mc^2}{mv}$$

$$\frac{E}{p} = \frac{c^2}{v}$$

$$v_p = \frac{c^2}{v_{particle}}$$

$$\boxed{v_p = \frac{c^2}{v_g}}$$ → quantum or light waves

matter waves

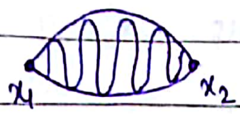
- We can't compare matter waves with light waves as characteristic of both of them are different.

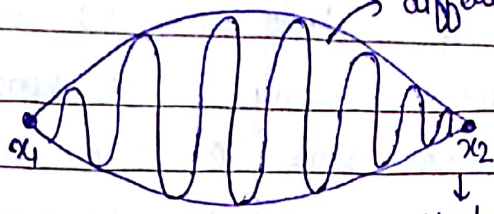
1) Energy-time 2) Position & Momentum
3) Polar coordinates

} Uncertainity Principle

Theoretical Derivation of Heisenberg Uncertainty principle.

Consider a small wave packet. Wavelength is very small in a small wave packet so momentum is max. So we can't measure momentum and position accurately and simultaneously.
having ω and wave vector K.





$$\Delta x \cdot \Delta p_x \simeq h$$

$$R = 2A \cdot \cos\left[\left(\frac{\Delta\omega}{2}\right)t - \left(\frac{\Delta K}{2}\right)x\right]$$

($\because$ bcoz at nodes resultant is zero.)

$$2A \cdot \cos\left[\left(\frac{\Delta\omega}{2}\right)t - \left(\frac{\Delta K}{2}\right)x_1\right] = 0 \quad -(1)$$

$$2A \cdot \cos\left[\left(\frac{\Delta\omega}{2}\right)t - \left(\frac{\Delta K}{2}\right)x_2\right] = 0 \quad -(2)$$

from eq. (1) $\Rightarrow \left(\frac{\Delta\omega}{2}\right)t - \left(\frac{\Delta K}{2}\right)x_1 = (2n+1)\frac{\pi}{2}$

from eq. (2) $\Rightarrow \left(\frac{\Delta\omega}{2}\right)t - \left(\frac{\Delta K}{2}\right)x_2 = (2n+1)\frac{\pi}{2} + \pi$ → phase difference b/w. particles in wave packet.

Subtracting the above equations:

$$\frac{\Delta K}{2}(x_2 - x_1) = \pi$$

$$\Delta x = x_2 - x_1 = \frac{2\pi}{\Delta K} \quad -(3)$$

We consider that particle is moving in straight line:

$$K = \frac{2\pi}{\lambda} = \frac{2\pi p_x}{h}$$

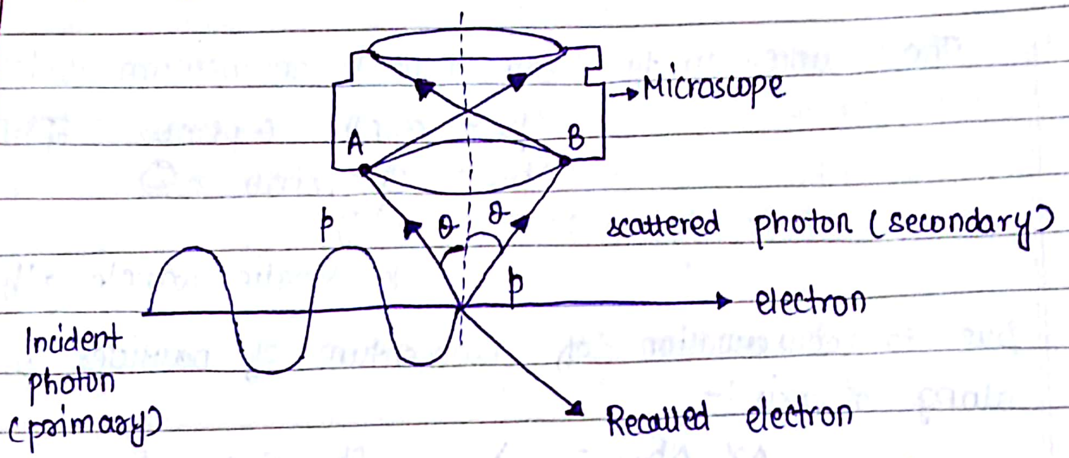
Small change in wave vector ← $\Delta K = 2\pi\left(\frac{\Delta p_x}{h}\right)$ — (4)

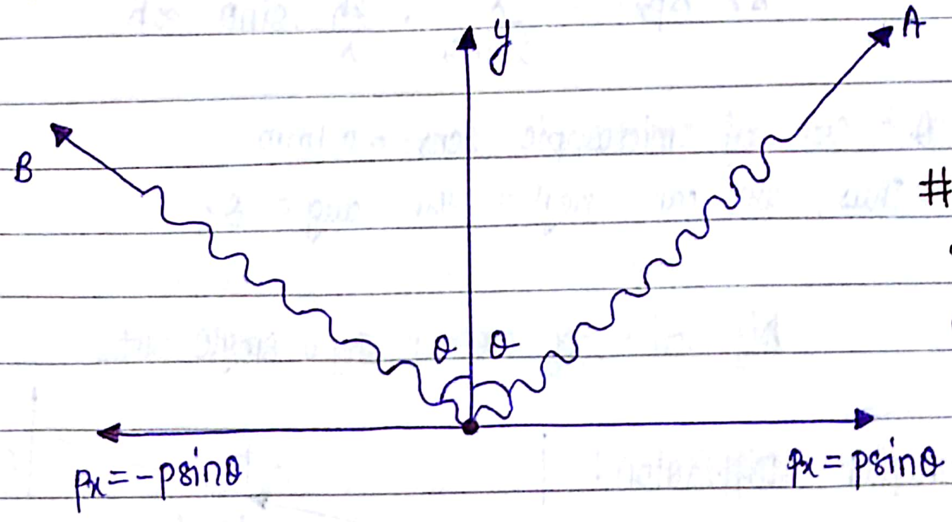
$$\Delta x = \frac{2\pi}{2\pi} \cdot \frac{h}{\Delta p_x}$$

$\boxed{\Delta x . \Delta p_x \approx h}$ hence proved.

Experimental proof of uncertainity Principle

(i) Determination of Position of particle by microscope; (Campten effect)





graph for movement of photons.

$\Delta x . \Delta p_x \approx h$ → The critical uncertainity Principle.

$\Delta E . \Delta t \approx h$
$\Delta \theta . \Delta \omega \approx h$ } cannot be determined simultaneously.

let us consider an incident photon at an electron and transfer its energy to the e^-.

$\Delta x \approx$ resolving power of microscope.

$$\Delta x \approx \frac{\lambda}{2\sin\theta} \quad -①$$

Δx → uncertainity in position

$2\sin\theta$ → The scattered photon moves in different direction.

The recorded e^- energy > energy of incident photon.

The uncertainity component in momentum is :-

$$\Delta p_x = p\sin\theta - (-p\sin\theta) = 2p\sin\theta$$

$$\Delta p_x = \frac{2h}{\lambda} \cdot \sin\theta \quad -②$$

λ → de-broglie wavelength

Due to conservation of momentum of particles is moving along x-axis :-

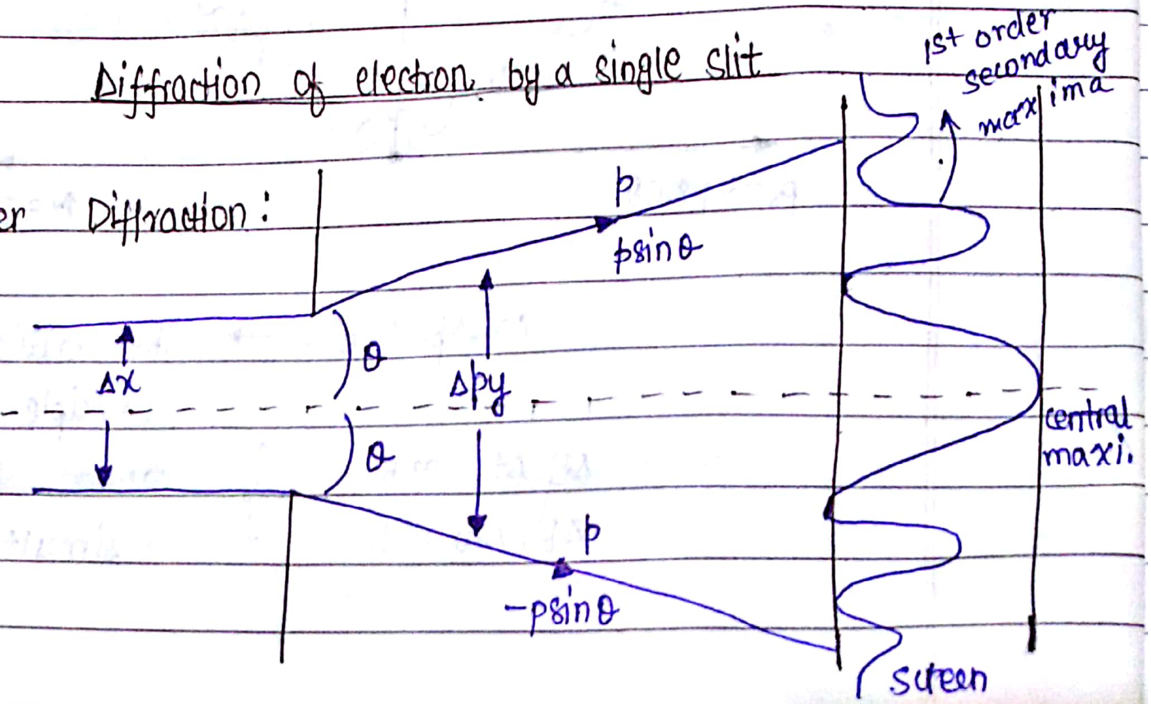
$$\Delta x \cdot \Delta p_x = \frac{\lambda}{2\sin\theta} \cdot \frac{2h}{\lambda} \cdot \sin\theta \approx h$$

θ = size of microscopic lens = 0.1mm

Thus, we can neglect the angle θ.

Diffraction of electron by a single slit

Fraunhoffer Diffraction:



Distance between screen and slit must be ∞.

$$2d\sin\theta = n\lambda \rightarrow \text{1st order secondary maxima}$$

2d is equivalent to Δx of uncertainity object.

$$\Delta x \sin\theta = n\lambda$$

$$n = 1$$

$$\Delta x = \frac{\lambda}{\sin\theta} \quad -(1)$$

The electrons are moving along x-axis they have no component of momentum along y-axis after diffraction they are deviated from their initial path and component of momentum lies anywhere between $p\sin\theta$ and $-p\sin\theta$. So, the uncertainity is defined by :-

$$\Delta p = p\sin\theta - (-p\sin\theta) = \frac{2h}{\lambda} \cdot \sin\theta$$

$$\Delta x \cdot \Delta p = \frac{\lambda}{\sin\theta} \cdot \frac{2h}{\lambda} \cdot \sin\theta \approx 2h$$

$$\approx h$$

$$\approx \hbar$$

Applications of Uncertainity Principle

1) Non-existence of electron in a nucleus :-

The energy associated with e^-

$$E^2_{min.} = p^2_{min.}c^2 + \underbrace{(m_0)^2 c^4}_{\text{constant (can be neglected.}} \longrightarrow \text{relation b/w. Energy and momentum}$$

$$E_{min.} = pc$$

The size of nucleus $\approx 10^{-14}$ m

Page No.
Date : / /

Q. Define wave-function ψ and its properties.

I) e^- exists in nucleus, Δx should be equal to 10^{-14}.

$$\Delta p_x = \frac{h}{2\pi . \Delta x} = \frac{6.63 \times 10^{-34}}{2 \times 3.14 \times 10^{-14}} \; J\text{-}s$$

$$= \frac{6.63}{2 \times 3.14} \times 10^{-20}$$

$$= 1.05 \times 10^{-20} \text{ kg m - sec.}$$

Energy of moving e^- = Beta decay $\approx$ 4-5 MeV (maximum energy)

$E_{max.} = 1.05 \times 10^{-20} \times 3 \times 10^{8} = 3.15 \times 10^{-12}$ J = 20 meV

Energy of e^- is very high compared to β-decay. Hence, e^- cannot exist inside a nucleus.

Wave-function : ψ varies in case of matter wave.

Probability density $|\psi|^2$ → probability of finding a particle in a particular time in a given wave packet.
↳ must be finite everywhere

$\psi = a + ib$
$\psi^* = a - ib$
$\psi . \psi^* = a^2 + b^2 = |\psi|^2$
↳ real and +ve

(i) ψ must be finite

(ii) ψ must be single value
probability at a particular point cannot be more than one

iii) ψ, $\frac{\delta \psi}{\delta x}$, $\frac{\delta^2 \psi}{\delta x^2}$ must be continuous everywhere
↳ probability can't exist at a discontinuous function

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Page No.
Date : / /

Condition of ψ in normalised and orthogonal wave :-

$$\int |\psi(x,y,z)|^2 \, d\tau = 1 \quad \rightarrow \text{normalised condition}$$

↳ space coordinates (# calculating ψ_n)

τ (Tau)

$$\int \psi_m^* \psi \, d\tau = 0 \qquad m \neq n \quad \rightarrow \text{Orthogonality}$$

↳ small volume

General Equation;

$$\int \psi_m^* \psi = \delta_{mn}$$

$\delta_{mn} = 0$ if $m \neq n$ → orthogonal
$\delta_{mn} = 1$ if $m = n$ → normalised

Schrodinger time Independent Equation.

$|\psi|^2 \rightarrow$ Probability Density.

$$\nabla^2 \psi = \frac{1}{u^2} \cdot \frac{\delta^2 \psi}{\delta t^2} \quad -(1)$$

$$\frac{\delta^2 \psi}{\delta x^2} + \frac{\delta^2 \psi}{\delta y^2} + \frac{\delta^2 \psi}{\delta z^2} = \frac{1}{u^2} \cdot \frac{\delta^2 \psi}{\delta t^2}$$

Solution of eq.(1) is $\Rightarrow \psi = \psi_0 \cdot e^{-i\omega t}$ —(2)

differentiate the equation w.r.t time :

$$\frac{\delta \psi}{\delta t} = \psi_0 \cdot e^{-i\omega t} \cdot (-i\omega)$$

$$\frac{\delta^2 \psi}{\delta t^2} = \psi_0 \cdot e^{-i\omega t} \cdot (-i\omega^2)$$

$$\boxed{\frac{\delta^2 \psi}{\delta t^2} = (-i\omega^2)\psi = -1 \times \omega^2 \psi}$$

$$\boxed{\nabla^2 \psi = -\frac{\omega^2}{u^2} \cdot \psi} \quad \text{as } \frac{\omega}{u} = k = \frac{2\pi}{\lambda}$$

Date : / /

$$\nabla^2\psi = -\frac{4\pi^2}{\lambda^2}.\psi$$

$$= -\frac{4\pi^2}{h^2} \times m^2 v^2 \psi \qquad \left(\because \hbar = \frac{h}{2\pi}\right)$$

$$\nabla^2\psi = \frac{-1}{(\hbar)^2} \times m^2 v^2 \psi$$

$$E = \frac{1}{2}mv^2 + V$$

$$2m(E-V) = m^2v^2$$

$$\nabla^2\psi = \frac{-1}{\hbar^2}.2m(E-V)\psi$$

$$\nabla^2\psi + \frac{8\pi^2 m}{h^2}(E-V)\psi = 0$$

or

$$\nabla^2\psi + \frac{2m}{\hbar^2}(E-V)\psi = 0$$

Schrodinger time Dependent Equation

$$\psi = \psi_0 . e^{-i\omega t}$$

$$\frac{d\psi}{dt} = \psi_0(-i\omega).e^{-i\omega t} = -i\omega\psi = -i \times 2\pi v \psi = -i \times \frac{2\pi E}{h}.\psi$$

$$E.\psi = i\hbar.\frac{\delta\psi}{\delta t}$$

$$\nabla^2\psi + \frac{2m}{\hbar^2}[E-V]\psi = 0$$

$$\nabla^2\psi + \frac{2m}{(\hbar)^2}\left[i.\hbar.\frac{d\psi}{dt} - V\psi\right] = 0$$

$$\nabla^2\psi = \frac{-2m}{(\hbar)^2}\left[i.\hbar.\frac{d\psi}{dt} - V\psi\right] = 0$$

$$-\nabla^2\psi \times \frac{(\hbar)^2}{2m} = i\hbar \cdot \frac{d\psi}{dt} - V\psi$$

$$\boxed{-\frac{(\hbar)^2}{2m} \cdot \nabla^2 \cdot \psi + V\psi = i\hbar \cdot \frac{d\psi}{dt}}$$ # Associated with time

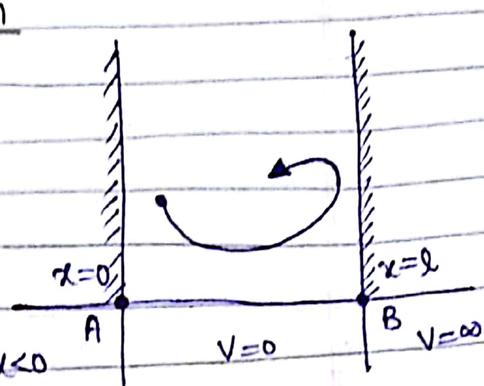
Applications of Schodinger Wave Equation

Particle in a box : $H\psi = E\psi$

$H \rightarrow$ Hamilton function

$V(x) = 0$

$= \infty$



$x<0$ and $x>a$

The box refers to a quantum system in which the potential energy of particle is zero within the region.

$\underbrace{X=0 \text{ and } X=l}_{\text{Boundary condition}}$ } at these points the probability of finding the particle is zero.

Time-Independent Schrodinger Wave Equation

Associated with Energy

$$\boxed{\nabla^2\psi + \frac{2m}{\hbar^2} \cdot (E-V)\psi = 0} \quad -(1)$$

(Potential energy of particle is zero within the system)

$$\nabla^2\psi + \frac{2m}{(\hbar)^2} \cdot E\psi = 0$$

let $K = \sqrt{\frac{2mE}{(\hbar)^2}} \Rightarrow \nabla^2\psi + K^2\psi = 0 \quad -(2)$

$$\frac{\delta^2\psi}{\delta x^2} + K^2\psi = 0$$

Sol.^{n} of eq. (2) is : $\boxed{\psi(x) = A\sin Kx + B\cos Kx} \quad (3)$

Applying Boundary Conditions:- $\psi(x)=0$ at $x=0$
$\psi(x)=0$ at $x=l$.

Put $x=0$ in eq.(B): $\psi(0)=0+B$
$\boxed{B=0}$

Put $x=l$ in eq.(B): $\psi(l)=A\sin(Kl)+0=0$

$\Rightarrow \sin\pi = A\sin(Kl)$

$\Rightarrow n\pi = Kl$

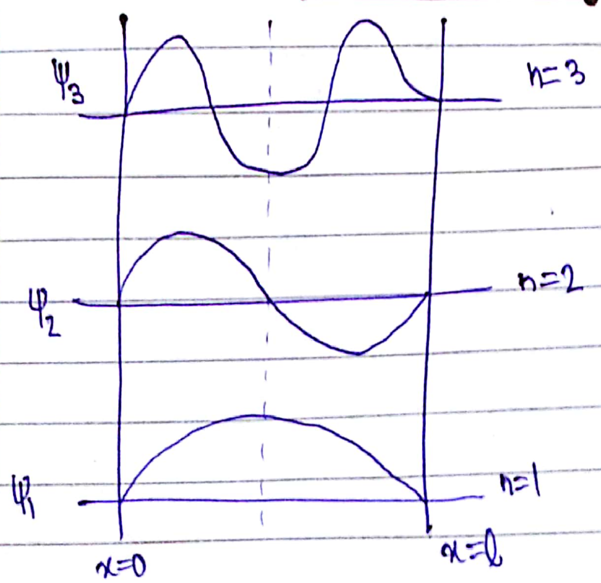
$\Rightarrow \boxed{K=\frac{n\pi}{l}}$

Thus, we have $\left(\frac{n\pi}{l}\right)^2 = \frac{2mE}{(\hbar)^2}$

Wave function associated with moving particle. $\psi = A\cdot\sin\left(\frac{n\pi x}{l}\right)$

Energy Eigen values $= \left(\frac{n\pi}{l}\right)^2\left(\frac{h}{2\pi}\right)^2 \times \frac{1}{2m}$

$$\boxed{E_n = \frac{n^2h^2}{8ml^2}} \qquad n=1,2,3,\ldots$$



Applications of Schrodinger Equation:-

(Calculate ψ value for max. conditions) ← $\boxed{\int_0^l |\psi|^2 . dx = 1}$ → $\int_0^l A^2 . \sin^2\left(\frac{n\pi x}{l}\right) . dx = 1$

On integrating we get → $\boxed{A = \sqrt{\frac{2}{l}}}$

$$\psi = \sqrt{\frac{2}{l}} . \sin\left(\frac{n\pi x}{l}\right)$$

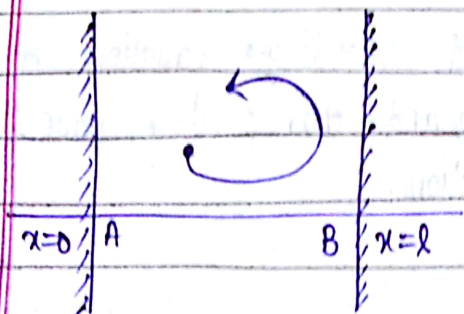
|02/18 **Zero - Point Energy:**

Wave function associated with moving particle : ψ_n

$$\psi_n = \sqrt{\frac{2}{l}} . \sin\left(\frac{n\pi x}{l}\right)$$

↳ length of box

Applying boundary conditions b/w. wall of a quantum system :-



If $n=0$, $E_n=0$, $\psi_n=0$.
↓
particle is in ground state

∴ Probability of finding particle in ground state is zero.

Thus, we can't apply Schrodinger time dependent and independent equations.

$n=1$, $E_n = \frac{h^2}{8ml^2}$, $\psi = \sqrt{\frac{2}{l}} . \sin\left(\frac{\pi x}{l}\right)$

at 0K in a quantum system

- Min. energy associated with a moving particle having $E_n = \frac{h^2}{8ml^2}$, $\psi = \sqrt{\frac{2}{l}} \sin \frac{\pi x}{l}$, is called Zero Point Energy.

Page No.
Date : / /

Postulates of quantum Mechanics:

1) For every observable physical quantity, there is a wave function defined by ψ.

2) Operators are used to change the mathematical functions in quantum Mechanics.

$$\text{Energy operator} = i\hbar \frac{\delta}{\delta t}$$

$$\text{Standard Momentum operator} = \frac{\hbar}{i} \nabla$$

Along Axes :- $p_x = \frac{\hbar}{i} \cdot \frac{\delta}{\delta x}$; $p_y = \frac{\hbar}{i} \cdot \frac{\delta}{\delta y}$; $p_z = \frac{\hbar}{i} \cdot \frac{\delta}{\delta z}$

$$\boxed{H\psi = E\psi}$$

Hamiltonian operator

Energy Operator

Eigen Value :

The values of energy for which schrodinger equation can be solved are called Eigen Values and corresponding wave function is called Energy Eigen Function.

Probability, $P = \int_0^{x/4} |\psi|^2 dx$